

Quantum Approach to Solids - Problem Sheet 1

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* indicates harder question.

1. Bonding

Which type of bonding is likely to be present in the following materials? Justify your answers.

- (a) argon
- (b) potassium iodide
- (c) silver
- (d) silicon

2. Unit cells and motifs

Consider the following pattern:

qp	db	qp	db	qp	db
db	qp	db	qp	db	qp
qp	db	qp	db	qp	db

Indicate:

- (a) a rectangular unit cell
- (b) a primitive unit cell
- (c) the basis of letters for each type of unit cell.

3. Packing fraction

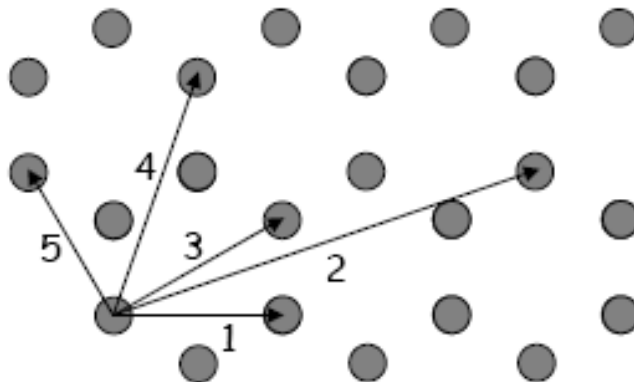
Calculate the packing fraction (i.e. the volume fraction occupied by atoms) for the following lattices, assuming that there is one atom on each lattice site, and that the atoms are hard spheres:

- (a) fcc, or face-centered cubic (also known as cubic close packing)
- (b) bcc, or body-centered cubic
- (c) hcp, or hexagonal close packed
- (d) simple cubic

* Find the primitive lattice vectors (and unit cells) for each of these lattices, and calculate the angles between the primitive lattice vectors in each case.

4. Graphite

The figure below is a representation of the in-plane crystal structure of graphite. Identify by number which, if any, of the marked vectors can be possible lattice translation vectors, and justify your answers. Sketch an example of the smallest unit cell and identify the basis.



5. Plane spacings

- (a) Sketch a simple cubic lattice, where the unit cell has side length a .
- (b) Mark in the planes: (001), (100), (101), (111), (211), (120).
- (c) Calculate the spacings between planes of these types (also known as d -spacings).
- (d) Show that the spacing of planes of the type (hkl) is:

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}.$$

NOTE: This formula is only true for cubic lattices.

6. Evaluating a crystal lattice

A crystal has a basis of one atom per lattice point and a set of translation vectors:

$$\mathbf{a} = 3\mathbf{i}; \quad \mathbf{b} = 3\mathbf{j}; \quad \mathbf{c} = 1.5(\mathbf{i} + \mathbf{j} + \mathbf{k});$$

where $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are unit vectors in the x , y and z directions of a Cartesian coordinate system. What is the Bravais lattice type of this crystal? What are the Miller indices of the set of planes most densely populated with atoms? Calculate the volumes of the given unit cell and the conventional unit cell for this lattice, assuming that a unit vector has length 1 Å.

7. The rocksalt structure

NaCl has a simple structure, which is often referred to as the rock salt structure. The Na and Cl ions are organized in interlocking fcc lattices. The lattice parameter is 5.63 Å. Give the Miller indices of the types of plane that consist of just one type of ion (i.e. all Na ions in one plane). These planes correspond to the ‘close-packed’ planes. Calculate the distance between:

- (a) close-packed Na planes;
- (b) close-packed Cl planes;
- (c) Na and Cl close-packed planes.

Calculate the density of salt in units of kg.m^{-3} .